Module 2: A+ Peripherals and Power Supply

DAY 2:-

Task 1:

Devices using SMPS (Switched Mode Power Supply):

1. **Desktop Compute**r : Needs stable, efficient DC power at multiple voltages (+5V, +12V, etc.) to run internal components like the motherboard and drives.

2. **Laptop charger/adapter** : Converts AC mains power to regulated DC efficiently while staying compact and lightweight for portability.

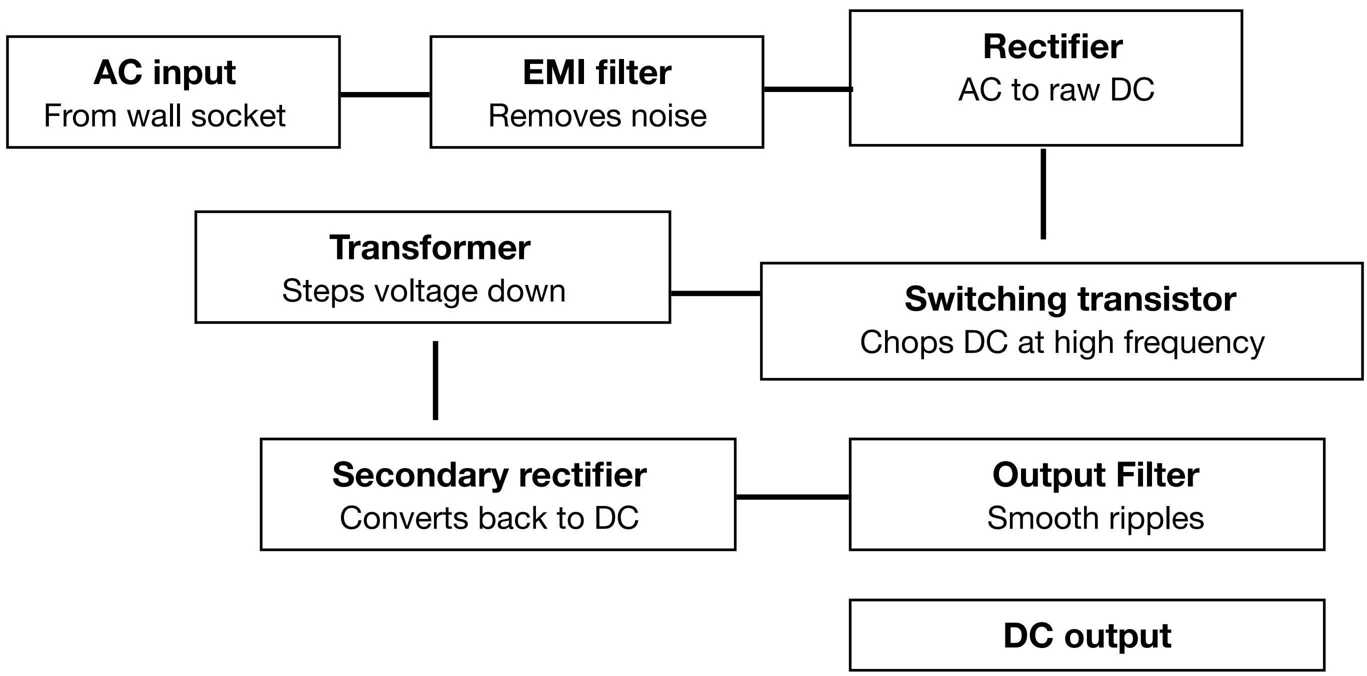
3. **LED TV/Monitor** : Requires efficient, low-heat power conversion to run display panels without bulky, heavy transformers.

Devices using UPS (Uninterruptible Power Supply):

1. **Desktop computer (in a college lab or office)** : Prevents data loss and file corruption from sudden power cuts by providing backup power to save work and shut down safely.

2. **Networking equipment (Wi-Fi router/server)** : Needs continuous, uninterrupted power to maintain internet connectivity and avoid disruption during outages.

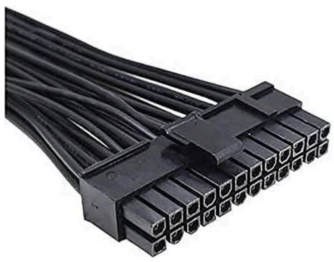
Task 2:



Task 3:

Power Connectors:

1. ATX-24 Power Connector



Powers: The motherboard itself - supplies power to the CPU socket, RAM slots, chipset, and onboard components.

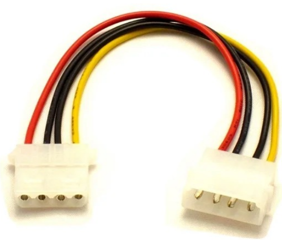
Safety tip: Always ensure the PC is fully powered off and unplugged from the wall before connecting or disconnecting it.

2. SATA power connector

Powers: SATA hard drives, SSDs, and optical (DVD/Blu-ray) drives.

Safety tip: Insert it gently in the correct orientation (it's keyed, so it only fits one way) — forcing it in upside down can bend or snap the thin plastic pins.

3. Molex connector



Powers: Older peripherals like case fans, fan controllers, LED strips, and some older optical/hard drives.

Safety tip: Hold the connector body (not the wires) while pulling it out — it can be stiff to remove, and yanking the cable can damage the internal wiring or terminals.

4.PCIe 6/8-pin connector

Powers: Dedicated graphics cards (GPUs) that need more power than the PCIe slot alone can supply.

Safety tip: Never force-plug it into the wrong orientation or into a similar-looking CPU connector — mismatched connectors can deliver incorrect voltage and damage the graphics card.

Task 4:

Step 1: Power off and unplug everything
- Before connecting anything, make sure the PC is unplugged from the wall socket and the SMPS switch is off.

Step 2: Connect the ATX 24-pin to the motherboard
- Connect the large 24-pin ATX connector from the SMPS into the matching 24-pin slot on the motherboard. Press down until the clip clicks in to lock it securely.

Step 3: Connect the CPU Power Connector
- Plug the smaller 4/8 pin connector from the SMPS into the CPU power socket near the top of the motherboard.

Step 4: Connect the GPU using PCIe 6/8- pin connectors
- Attach one or two PCIe 6/8-pin connectors from SMPS directly into the power sockets on the graphics card.

Step 5: Connect storage devices using SATA power connectors
- Plug a SATA power connector from the SMPS into each SSD or hard drive’s power port.

Step 6: Power on and test the system
- Plug the PC back into the wall, turn on the SMPS switch and press the power button to confirm the system boots correctly with all components recognized.